

Goal

The goal of the notebook is to analyse the steam market of games. Target Variables of this project are **the number of copies sold and amount of people playing the game**

The analysis will be based on the features such as:

- core features(owners, players, reviews)
- tags
- genres
- languages
- price
- reviews

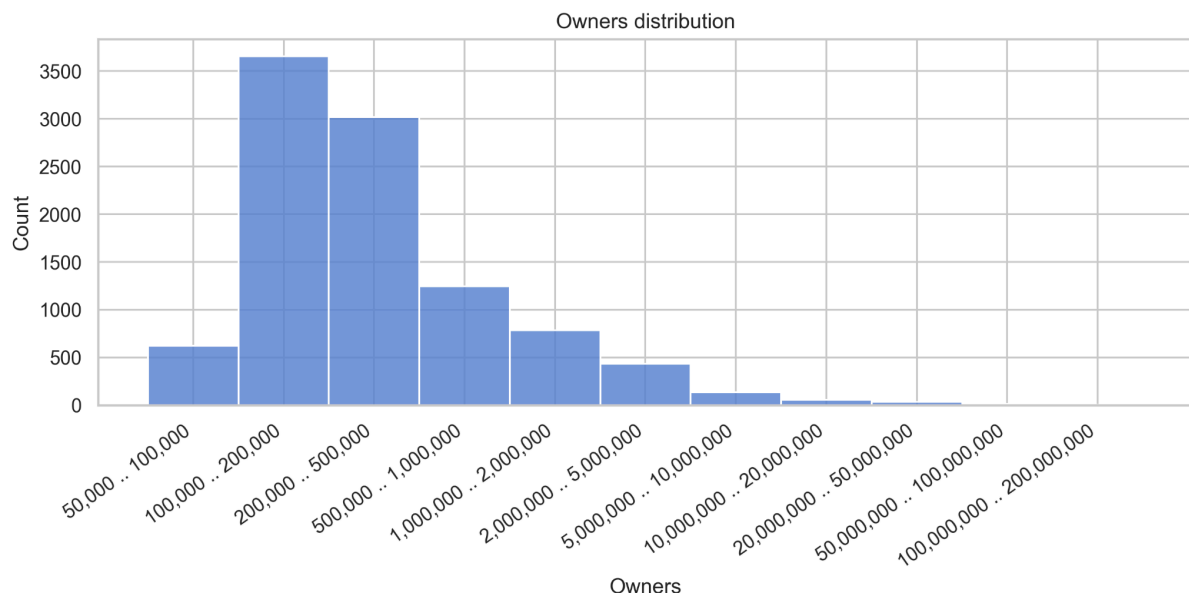
The dataset consists of the top 10000 games on steam ordered by the amount of owners. All the data was gathered with SteamSpy API on November 10th 2025.

Copies sold

In this section we will analyse how other features affect the amount of people who own the game. It includes people who got the game for free, via keys or with discounts.

The values that we are going to analyse are 'owners' and 'owners_numeric'. 'owners' is an ordinal value which represents the bin in which the number of people owning the game falls in. It is a string that looks like "lower_border .. upper_border".

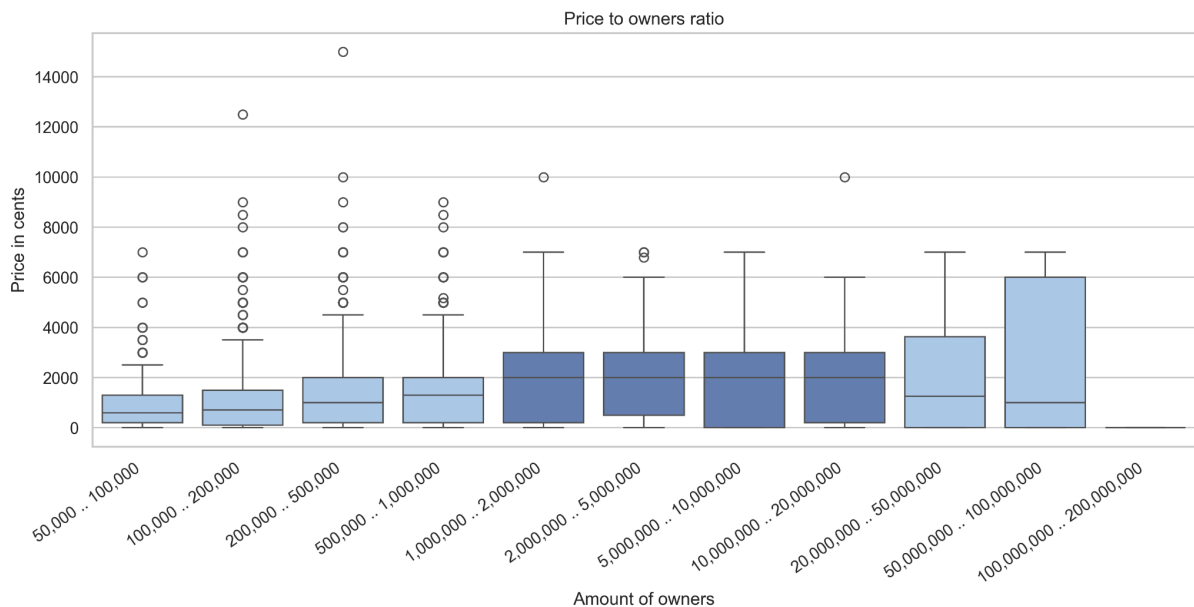
'owner_numeric' is an integer that represents the mean of the upper and lower borders of the bin. We will mostly work with 'owners' as for 'owners_numeric' was created as an estimation for mathematical purposes.



At first glance this distribution looks like a right-skewed distribution. However, this shape is affected by the fact that the sample contains only top 10000 games ordered by the number of owners. This way we don't have most of the games that have less than 100,000 owners. In reality the population **probably follows a heavily-tailed distribution** such as power-law.

Correlation with other variables

Price

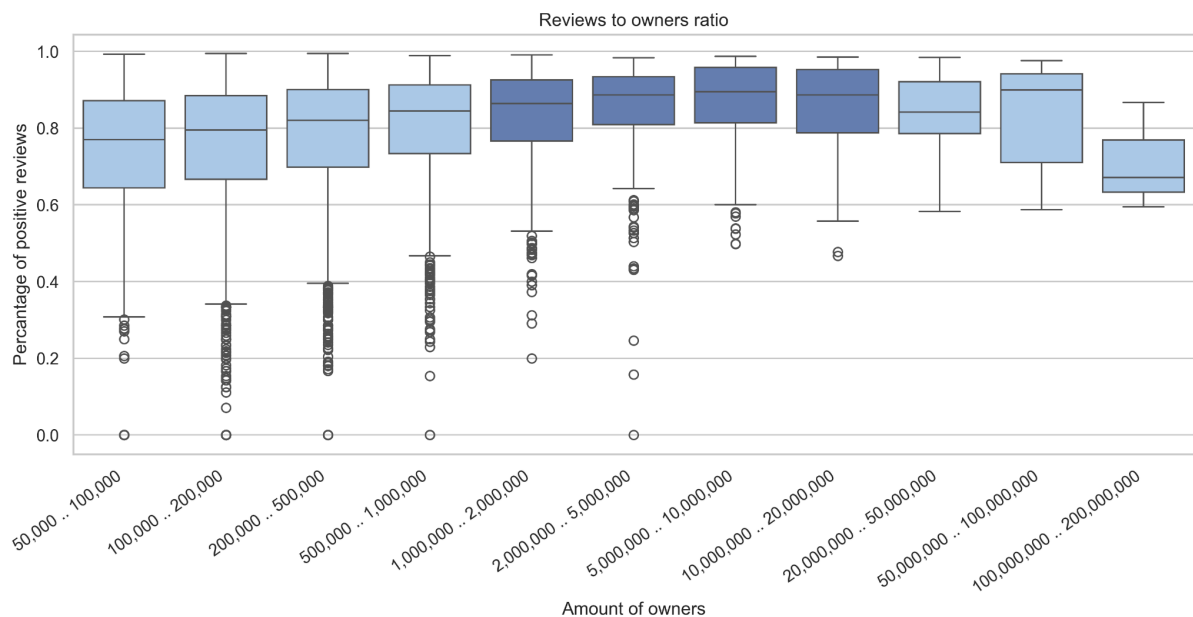


From this plot we can see that **there is no one true strategy to choose a price.** But we still can see a few patterns.

- The median of the price tends to go higher with the amount of owners until it reaches 1,000,000. From there till 20,000,000 it stays at median of 20 dollars. This suggests that games from this range mostly have a good quality to price ratio. From 20,000,000 and more median tends to fall until the bin with highest ownership where all games are free. It is completely expected for free or cheap games to have more players.
- Games from the range 100,000,000 to 200,000,000 are: *"Counter-Strike: Global Offensive"*, *"Apex Legends"*, *"PUBG: BATTLEGROUNDS"*. **All of which are free-to-play online shooters. Which means these games tend to gather a big audience around them.**
- Bin with 50,000,000 to 100,000,000 owners has really huge variance, despite a low median. Reason behind this are AAA games made by large studios with priceful games but a big fanbase.

Next we are going to test the theory if games with the number of owners ranging from 1,000,000 to 20,000,000 actually do have a good quality.

Review ratio



The hypothesis about games between 1 million and 20 million owners having the best quality and price is supported by this plot. This group of games has a really high median of review ratio with a fairly small interquartile range, and the first quartile barely going below 80% in some cases. From this we can state that **games with 1,000,000 to 20,000,000 owners tend to offer the best deals.**

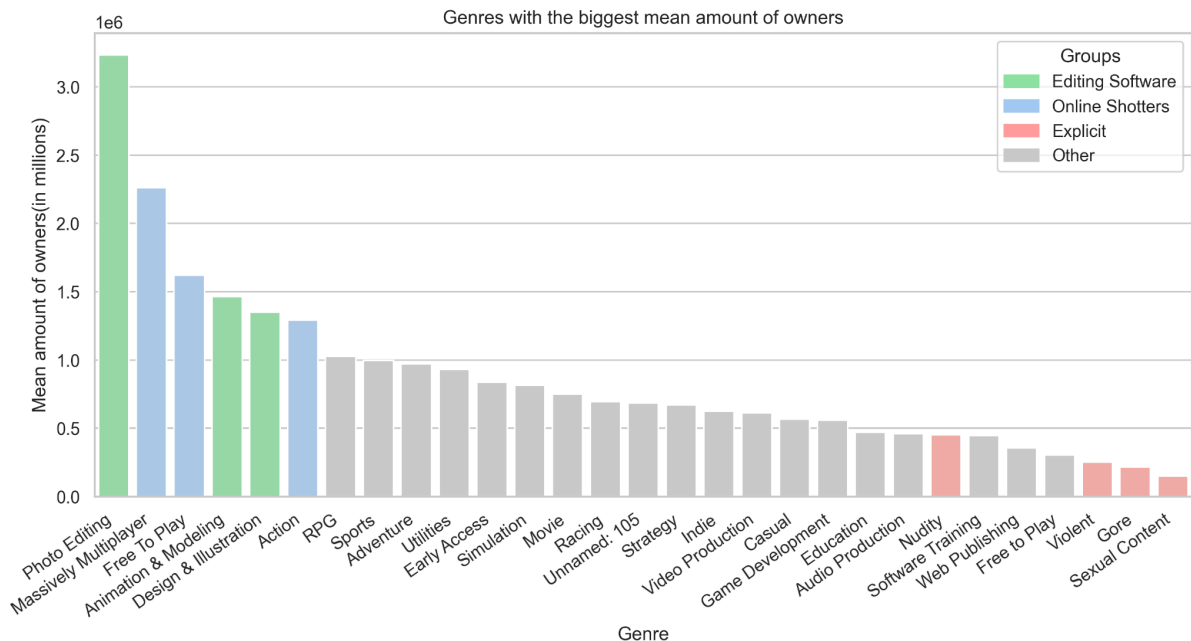
From this plot we also see an interesting situation with games that have from 50,000,000 to 100,000,000 owners. It naturally has a highest median but high variance is less expected. Cause of both facts could be AAA games once again.

Good projects with high quality raise the median, when failed projects from big studios with high price lower the first quartile.

Even more captivating is the case with the highest number of owners. It has the lowest median. It is hard to talk generally about it, since there are only three games in this bin. We can assume that it comes from the fact that those games are free-to-play. It forces developers to add microtransactions, which are generally hated in the gaming community.

Owners and categorical values

Popularity of genres

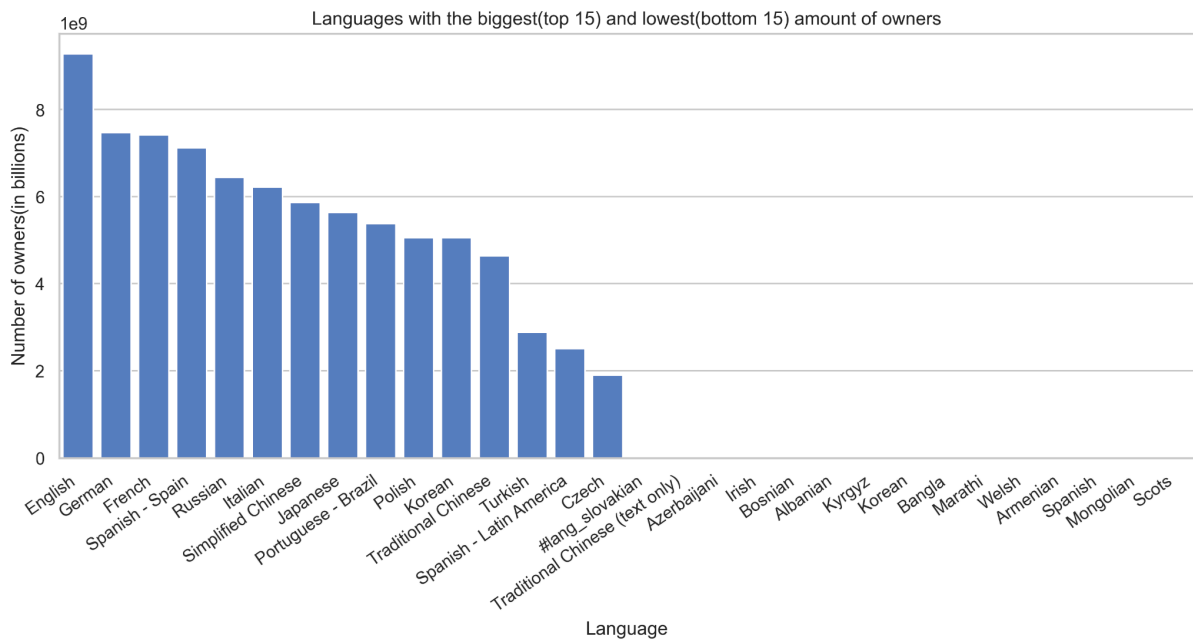


From this graph we already can see that there are 6 genres that dominate over other(having more than 1,000,000 owners on average). But they can be separated into two **the most popular categories**:

- **Editing softwares.** Their mean is being strongly carried by a couple applications that dominate in this field(like "*Blender*" or "*Wallpaper engine*"). There is not a big amount of these software on steam which makes the mean go even higher.
- **Online shooters.** This plot supports our observation by showing that free-to-play, online, action games do tend to get more players. But we have to notice that multiplayer and free games are not strictly connected to being action games. There are a lot of other popular games in these fields but we can not avoid the fact that it correlates with our previous statement.

On the other side we can see three genres with lowest averages. It is hard not to notice that they all fall into the category "explicit". Those games were not made for a large audience and never gained one.

What regions brings the most players

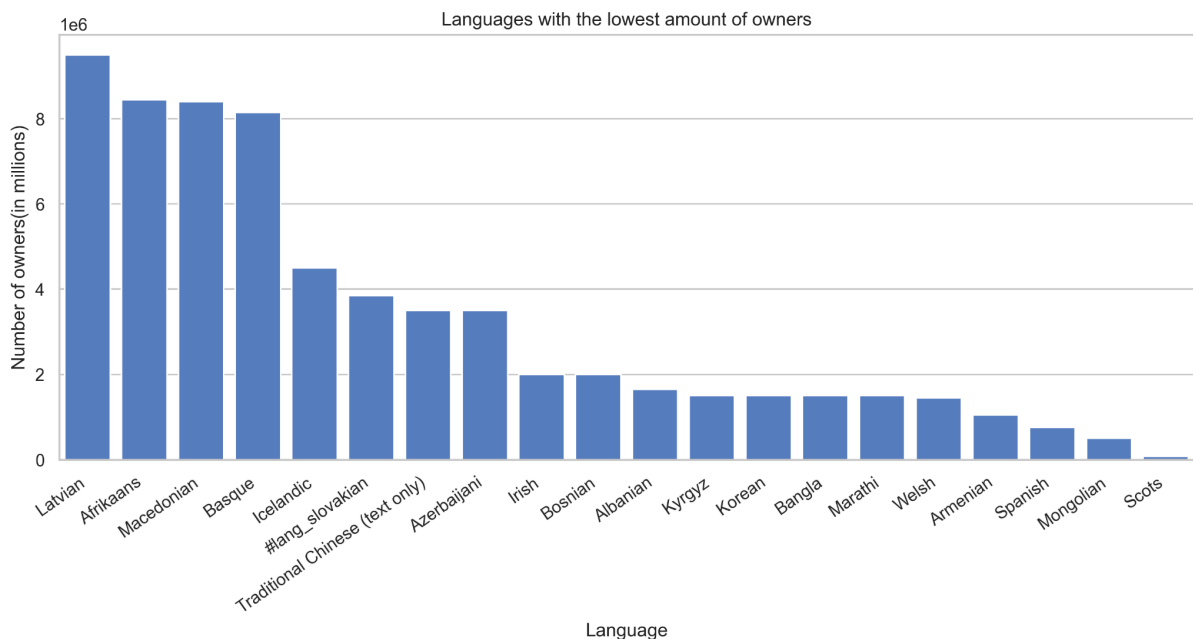


It is really hard to interpret this data. It can not be seen as a visualisation of how many players having an additional language brings to the game because of few reasons.

First of all, almost every game has English, German, French and Spanish since they are the most widely known languages.

Secondly, a game could have English and German. But even if everyone playing would only know English, German still would gain as much value as English.

Above all that, there is too big of a difference between the highest values and the lowest, making the lowest unreadable.



From this we can fetch at least some kind of insights. Such a low amount of value would mean that these languages are mostly present in niche games, and are not a good choice for growth. **Languages that appear in small-market games:**

- **Languages with a tiny player pool** such as Armenian, Mongolian, Kyrgyz, Bosnian.
- **Languages suppressed by other languages.** For example: Irish, Welsh, Marathi.

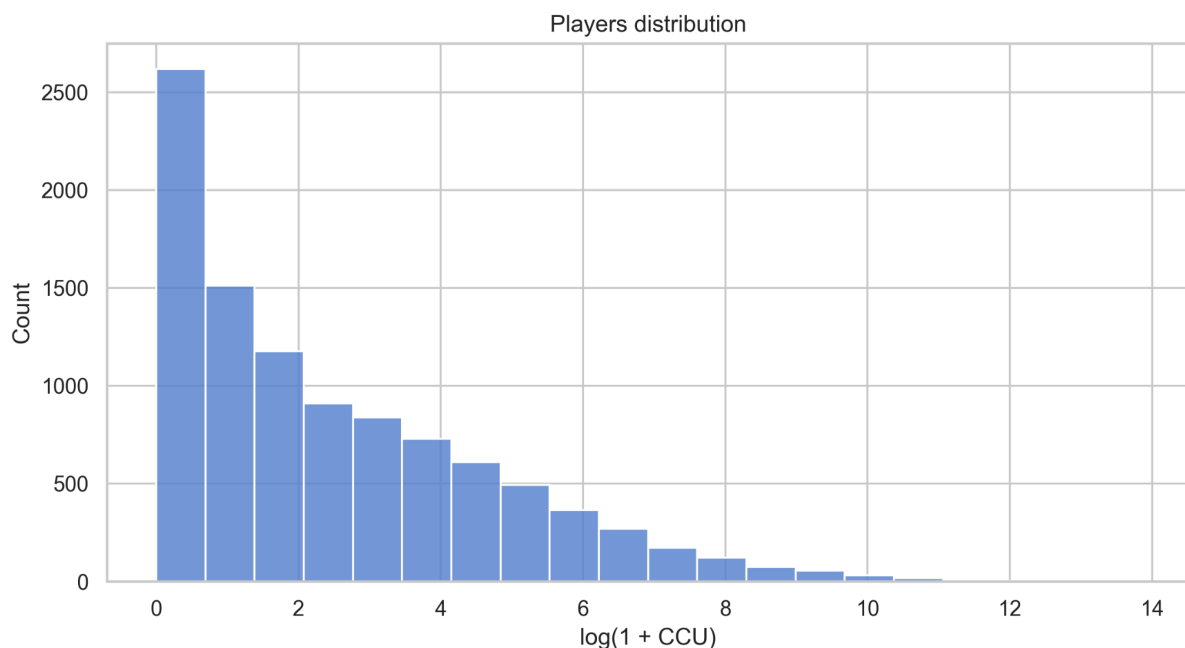
There is also noise. "Spanish" is an incorrect name because in steam, Spanish always comes with a name of a dialect(Spanish - Latin America).

Players online

In this section we will analyse what types of game were popular at the time data was gathered.

The main variable we are going to analyse in this section is 'ccu' which is an abbreviation for "concurrent players". It is a discrete variable which describes the number of players yesterday at the highest point(9th of November 2025). What's important to remember is that this type of data is really sensitive to the time when the data was gathered.

Luckily, we have other, more stable features like: 'average_2weeks', 'median_2weeks', 'average_forever' and 'median_forever'. They describe the average/median time a person spent in a game in a certain range of time. It is different from "concurrent players" since a game with a small number of owners still can have a high average playtime.

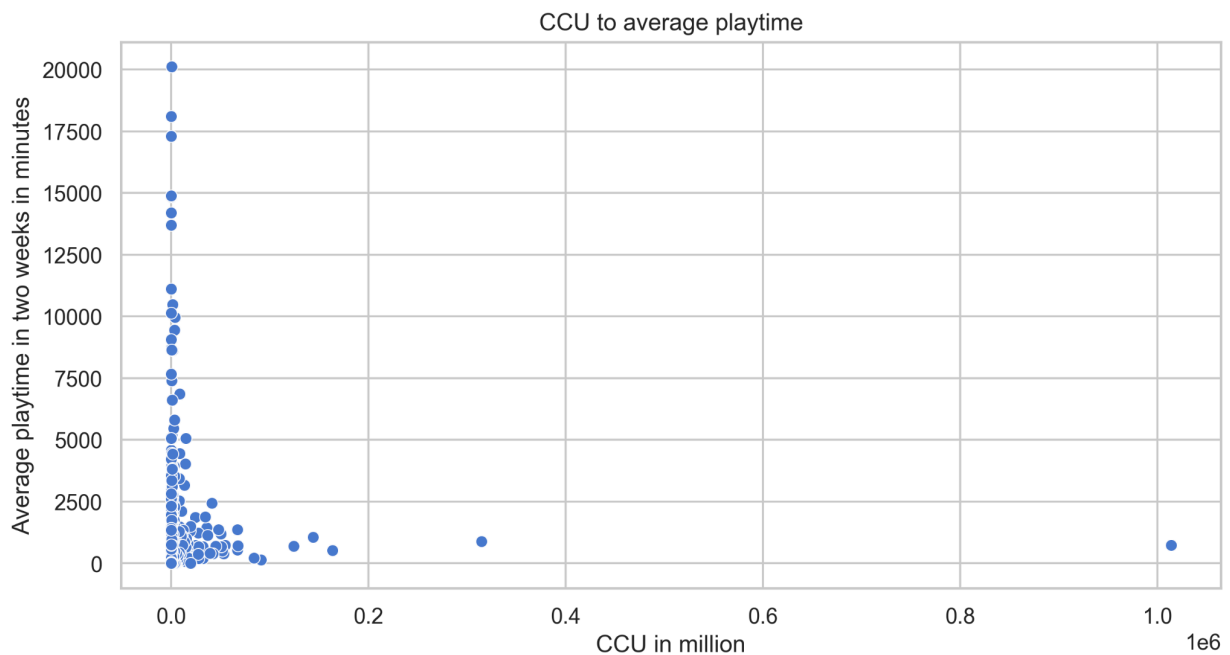


The problem with our data is the amount of games that have not been played recently. **Around 26% of games had no players recently and 57% of games had less than 10 players.**

Because of this log transformation ($\log(x + 1)$) was used for better visualisation. From both visualization and the number of games with a small in game online we can make a deduction that **the data is heavily-tailed.**

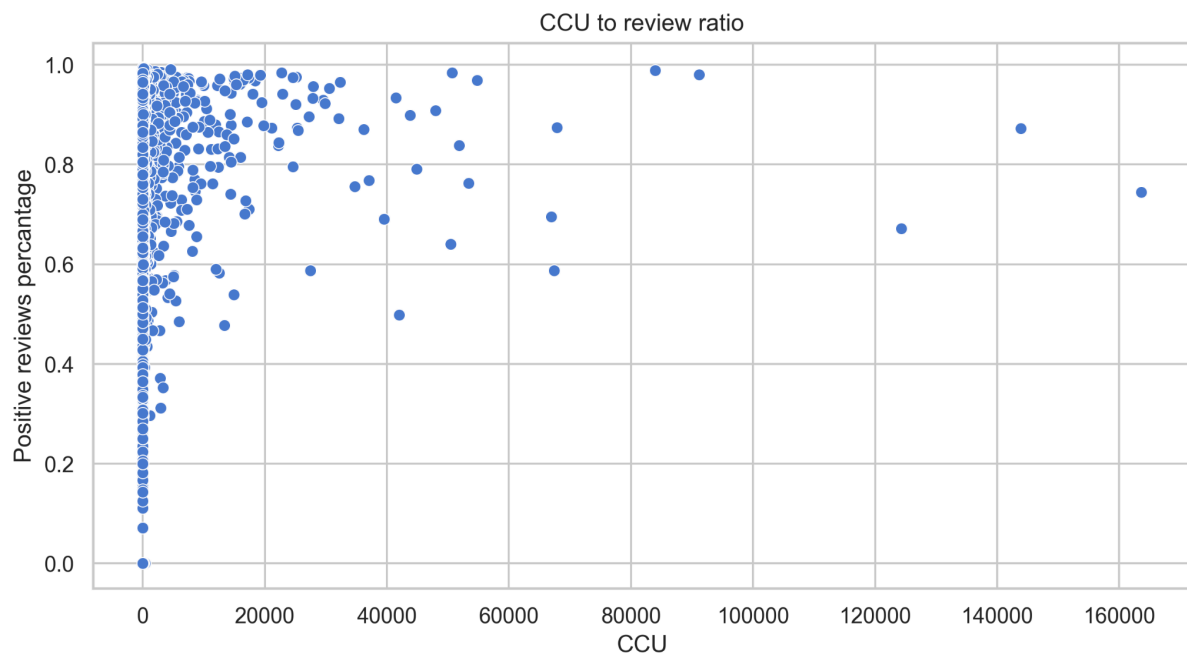
Correlation with numeric features

Average time in a game in two weeks



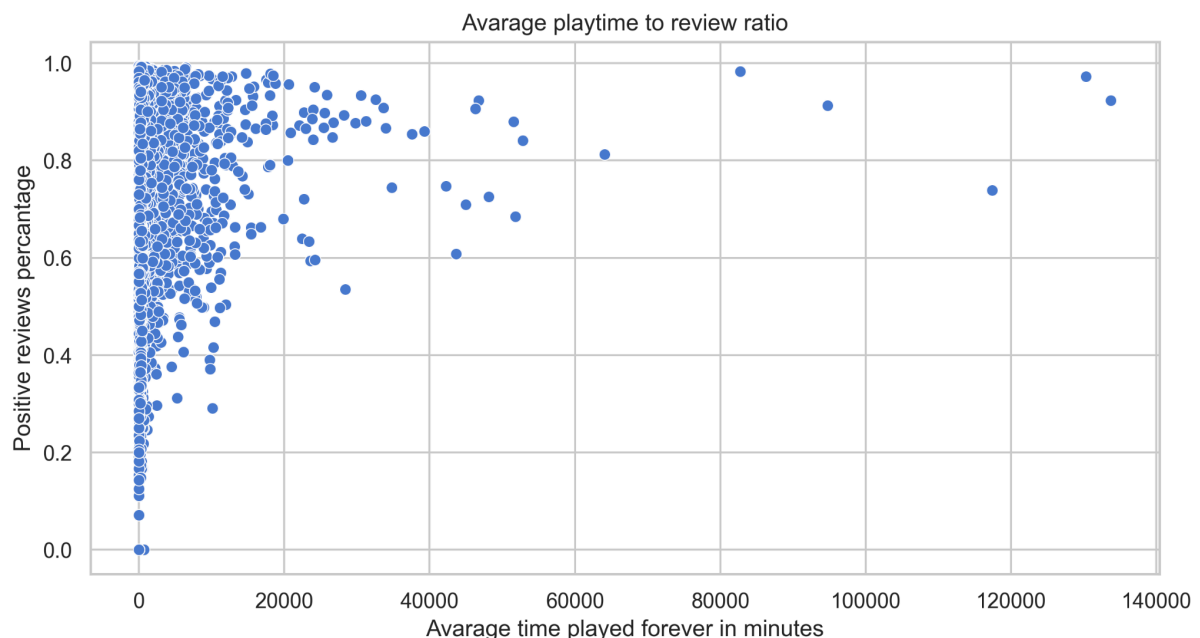
The scatter plot doesn't show strong positive correlation. Most of the points are clustered around low values of both concurrent users and average playtime. There is also a big variety of playtime values around for games with near zero players, meaning **small games still can have people spending a lot of time on them**. Furthermore, games with very high online tend to have relatively moderate average time in game, meaning **big projects bring big amount of casual players**, who do not tend to spend a lot of time in games.

Percentage of positive reviews



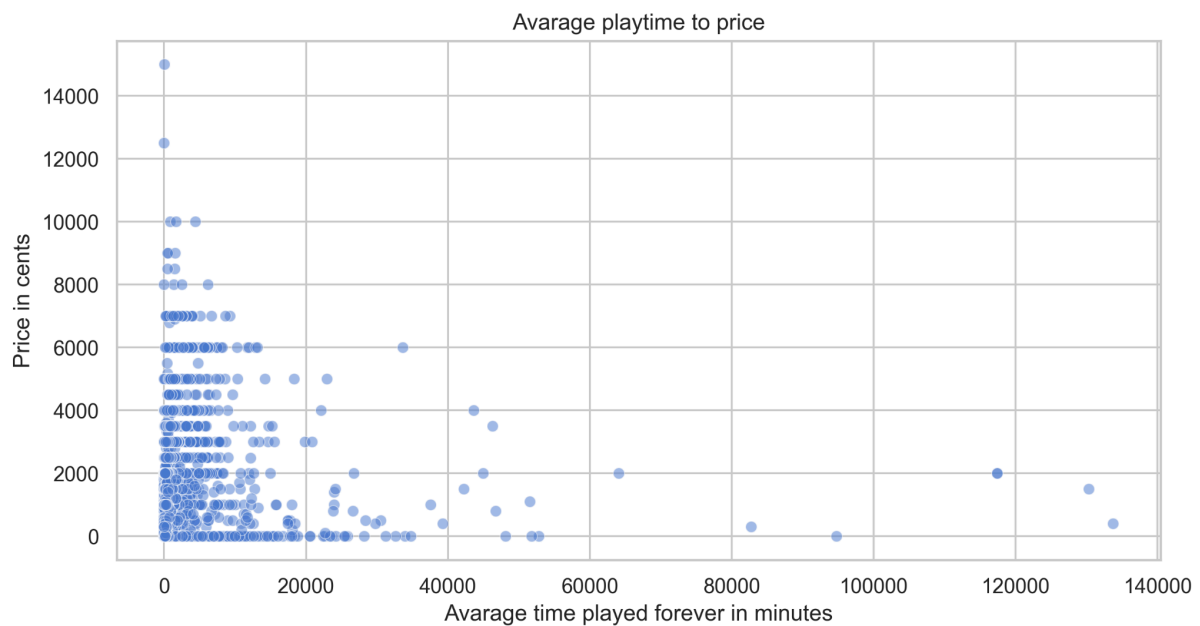
This scatter plot shows a slight correlation between ccu and review sentiment on the game. **There are only few games with less than 40% of positive opinion and having any people online.** After that there is a huge variance of values and no clear connection between amount of players and opinion.

For better readability two games with highest ccu has been eliminated.



Situation gets much clearer when we plot average playtime instead of players online. Correlation becomes more noticeable, meaning that **games with a lot of content and replayability tend to get better opinions**, and, logically, people play games they like more.

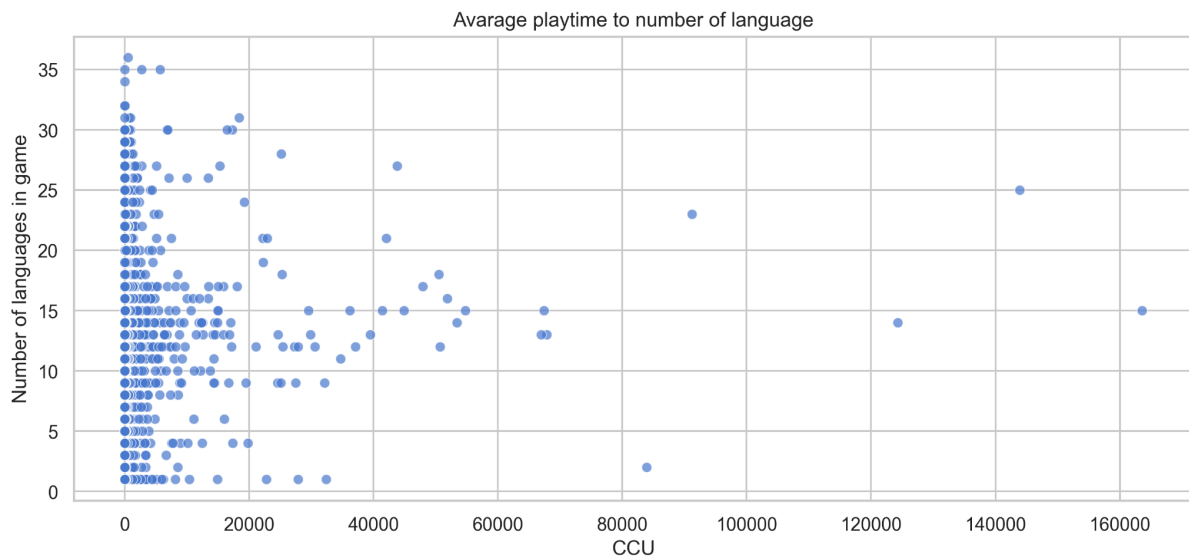
Price vs time spent in game



Interestingly this plot shows that people do not choose to spend a lot of time on costly games. Games with the highest average playtime per player have price no higher than 20 dollars.

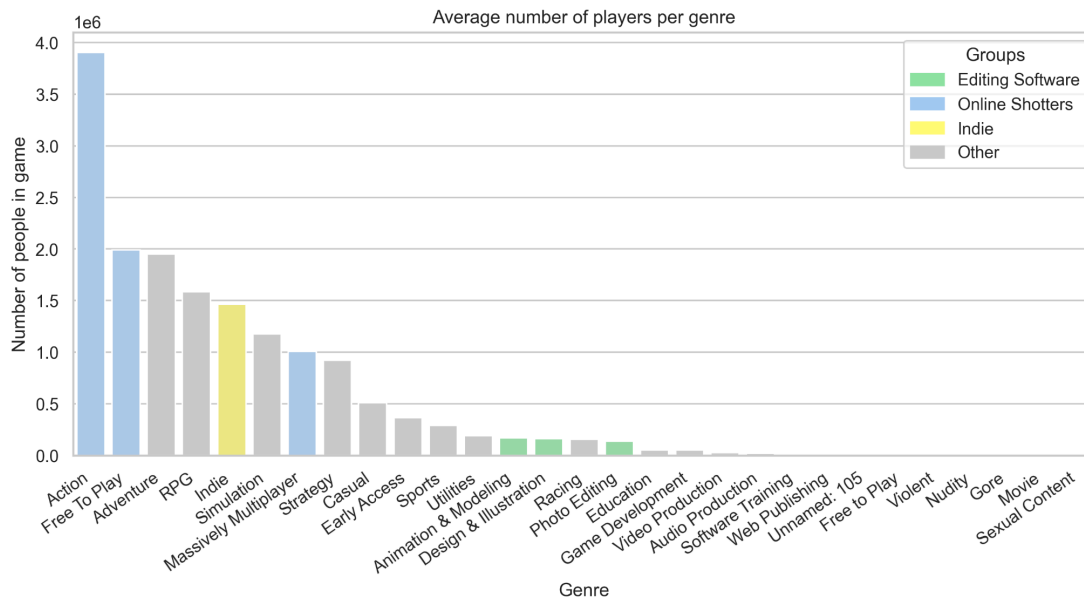
However, if we take a deeper look we can conclude that **a lot of games with high average playtime are being carried by a few players, who spend a lot of time in the program.** In reality, the median of those games is mostly less than 350 minutes. Nevertheless the game with the highest average still has a very high median (around 60,000). It is a cursor changer application that probably has to be open all the time for it to work.

Languages



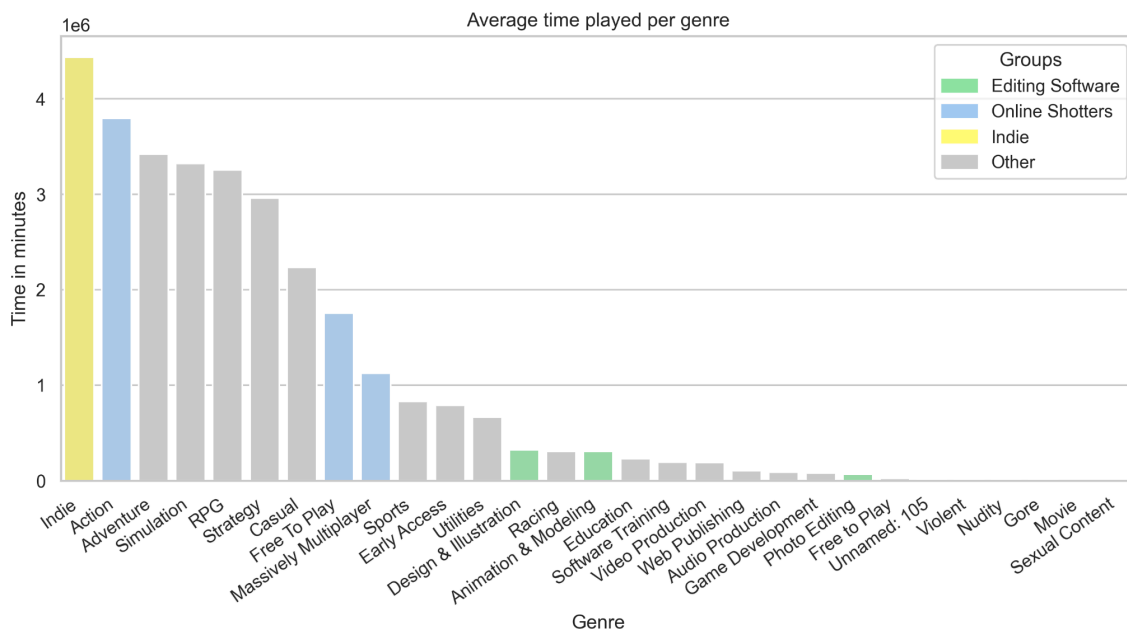
Categorical values

Genre



This plot shows certain similarities to the one we did previously with a number of owners. However, when we plot with concurrent users we don't have editing softwares at the top. Also one of the biggest changes is "Indie" tag. At the bottom the only noticeable difference is the tag "Movie". From this we can make few insights:

- A lot of people get editing softwares but don't use them regularly.
- Games from small teams gain popularity lately.
- "Movie" games lost their popularity.

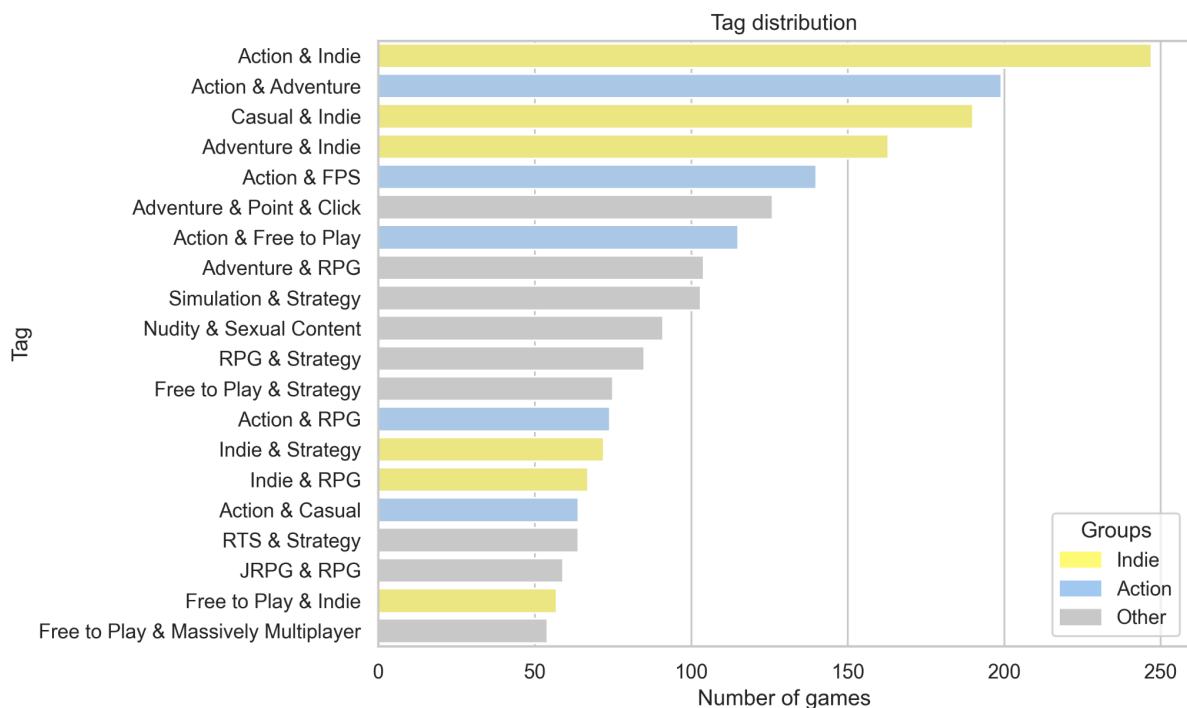


The plot shows, that not only indie games got more popular but also **people on average spend the most time in indie games**. Beside that we can conclude that free-to-play games can lure players into playing them, but they don't hold their audience for long time.

Tags

Tags are one of the most important features that we can analyse. It strongly impacts the target audience, highly affected by ongoing trends and one of the crucial factors for consumers.

On Steam, tag is being chosen by user votes. Our dataset contains two most popular options, represented in two columns with categorical values.

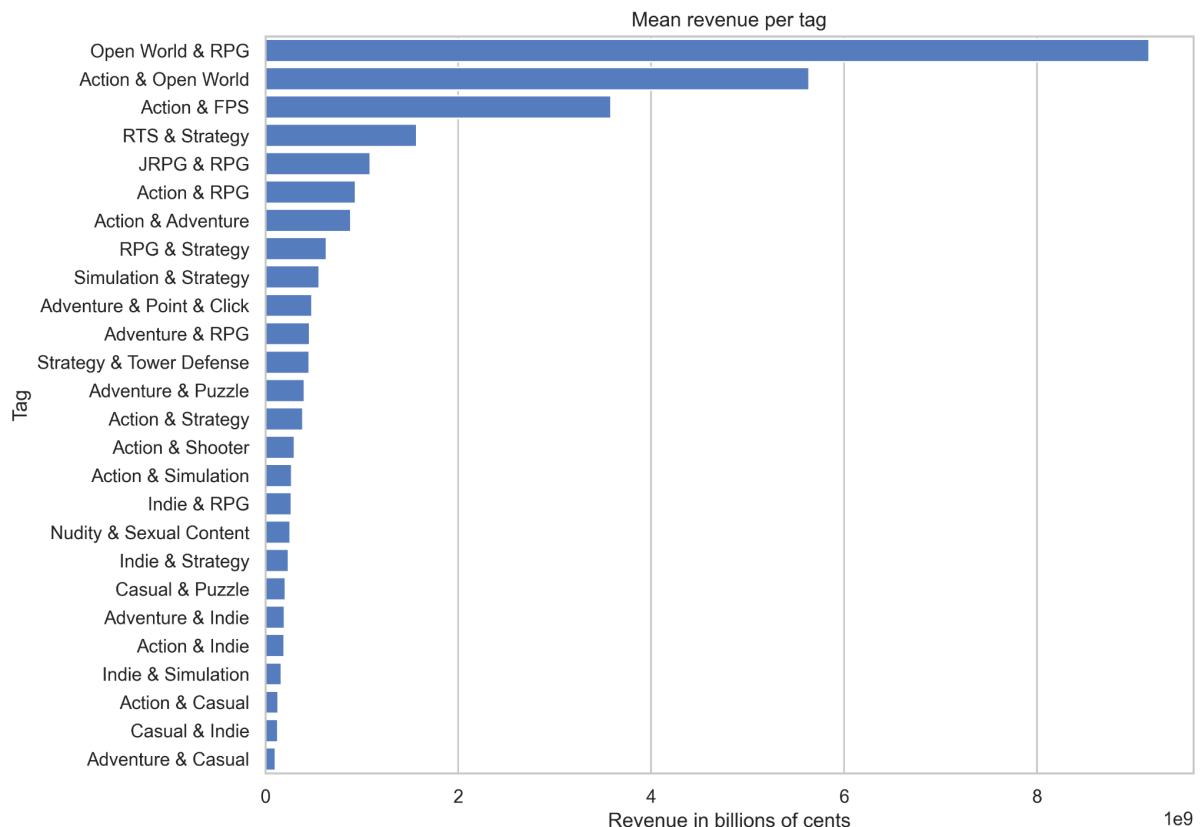


Presented visualisation shows the most numerous types of games. To prevent repetitions (such as "Action & FPS" and "FPS & Action" being two separate types of games) the tags have been sorted in alphabetical order before connection.

The most important insights:

- **The dominating tag is "Action & Indie"**, there are multiple reasons that could explain this, such as having low entry level for players or easy to make for developers.
- **Almost all top combinations have an Indie tag.** Those games don't need as many developers, money and time to make, making them much more common.
- **There is a lot of competition in the field of 'Action' games.**

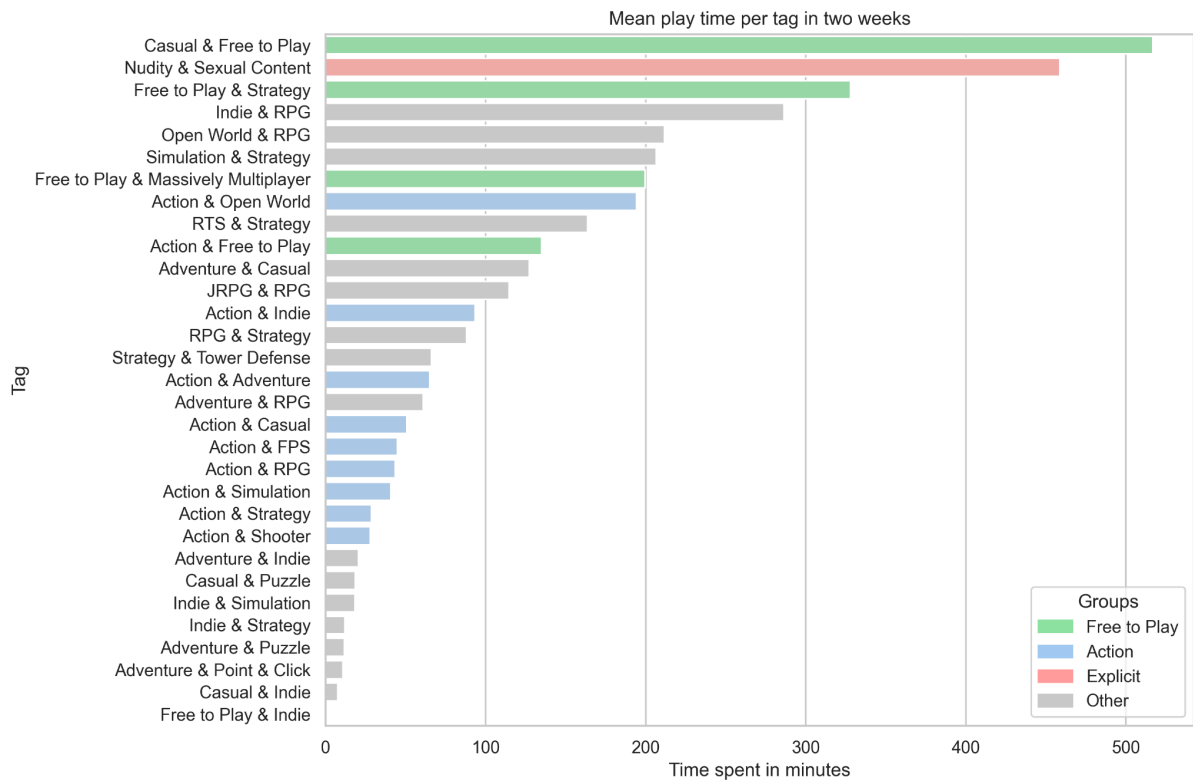
Biggest earns



This is one of the most interesting plots so far. It clearly shows what types of games have the biggest return in money from sales. Nevertheless, there are few details we have to remember before coming to conclusions. The revenue is estimated by the amount of people who bought the game times initial price. Therefore, it ignores the sales and microtransactions. Also for better readability and confidence we only used tag combinations with more than 40 games. What does the plot tell us?

- **The "Big Three" Dominance:** Combinations like **"Open World & RPG"**, **"Action & Open World"**, and **"Action & FPS"** are the clear leaders, despite not being the numerous. For example, the success of "Open World & RPG" lies behind big projects such as *"Skyrim"*, *"The Witcher"* and *"Assassin's Creed"*.
- It is important to note that "Action & Indie", which was the most frequent combination in our previous chart, appears near the bottom of the revenue list. This highlights that a **high number of games in a given category does not necessarily imply high sales returns**.
- Categories like "RTS & Strategy" and "JRPG & RPG" show solid performance. **While they have fewer titles, they show a loyal, paying audience, making them "premium" niches.**

Time spent



- **There is a clear cluster of action games around 30 to 60 minutes of playtime.** While being one of the most numerous tags people don't really choose it.
- **Free-to-play dominates engagement.** By using grind systems, competitive or persistent mechanics these games engage large groups of people in a strong short time involvement.
- The most unpredictable insight is the popularity of games with nudity. **Despite bad results in previous graphs people spend a lot of time in games with sexual content.** The reason can be a stable but small audience that spends a lot of time in the game.

Summary

- **Distribution of games by the number of owners follows the heavy tailed distribution.** Though we can not be certain about extremely unpopular games due to the size of the dataset. Nevertheless it means that most of the games don't gain popularity.
- **One of the most popular type of games by downloads are online first-person shooters, occupying 4 positions out of the top 5.**
- Seemingly, editing softwares also have a lot of users, but realistically the data is probably skewed by few programs such as "*Blender*" or "*Wallpaper engine*".
- Languages that are suppressed by English or have a small number of native speakers don't bring a big audience to the games.
- **Around 26% of games had no players recently and 57% of games had less than 10 players.** Even so it is still affected by the survivorship bias, since we don't look at the less popular games.
- The most successful games show that **it is more optimal to have a stable casual audience, rather than having a small group of dedicated players that play for a longer time throughout the day.**
- **Additionally games with smaller prices tend to have more players playing them.**
- Games with 10 to 25 game languages have more players playing them on average.
- **The most profitable tags are "Open World & RPG", "Action & Open World", and "Action & FPS"**
- Action games are numerous but most of the time people don't spend a lot of time on them.
- Games with sexual content show low performance on most of the parameters but they have a stable fanbase, leading them to have one of the highest mean time spent in two weeks.
- **Free-to-play dominates in engagement.** By using grind systems, competitive or persistent mechanics these games engage large groups of people in a strong short time involvement.